

TRAN HAI QUAN

AI Engineer & Researcher

Hanoi, Vietnam • (+84) 0334683088 • tranhaiquan16102006@email.com • github.com/pieupieu16

PROFESSIONAL SUMMARY

Passionate Artificial Intelligence student with strong practical foundations in Machine Learning, Multimodal AI, Retrieval-Augmented Generation (RAG), and Agentic System Architectures. Proven track record in designing cost-optimized multi-tier AI routing architectures (local ONNX edge inference + cloud VLMs), developing robust FastAPI/LangGraph backend microservices, and integrating IoT sensory data for automated logistics. Eager to contribute disciplined analytical capabilities, rapid prototyping, model experimentation, and production AI engineering teams.

FEATURED PROJECTS & ENGINEERING EXPERIENCE

GreenBin AI — Smart Waste Sorting & Recycling Logistics Agent | Core AI & System Developer 2026

Project Overview: *An end-to-end AIoT operational platform designed for Vietnam's Environmental Protection Law 2020 compliance, combining multi-tier multimodal classification with dynamic collection routing across residents, cleaners, and building managers.*

- **4-Tier Cost-Optimized Vision Routing:** Architected a cascading classification pipeline: **T0 (pHash cache, $0 cost)** → **T0.5 (Local CLIP ONNX int8, 56ms latency, 88.7MB)** → **T1/T2 Cloud VLMs (NVIDIA NIM / Gemini 1.5 Flash)**, slashing cloud API overhead by >60% while maintaining high recognition precision.
- **Agentic LangGraph Workflow & HITL Guardrails:** Implemented autonomous decision graphs with 3 Human-In-The-Loop (HITL) checkpoints for hazardous waste escalation, oversize load auditing, and cleaner logistics route approval.
- **Hybrid RAG Policy Retrieval:** Built a hybrid BM25 + Gemini Dense Vector (768-dim) retrieval system for building-level municipal regulations, achieving benchmark results of **hit@5 = 1.000** and **MRR = 0.838**.
- **Full-Stack Architecture & IoT Telemetry:** Developed the BQL administration portal (Next.js App Router, Tailwind) and integrated ultrasonic IoT bin fill-level telemetry into FastAPI backend with Supabase PostgreSQL and private cloud storage.

PIEUPIEU Review Care — RAG Customer Sentiment Pipeline | Core AI Developer 2026

Project Overview: *Automated e-commerce customer sentiment analysis and policy-compliant AI customer support pipeline delivering contextual responses with strict domain guardrails.*

- Built a high-performance RAG system utilizing **FastAPI, Streamlit, ChromaDB**, and an asynchronous **Redis Queue** microservices architecture.
- Integrated Gemini API for automated responses, boosting precision via a hybrid embedding-reranker pipeline (**multilingual-e5-large** + **bge-reranker-base**) for policy-grounded generation.
- Deployed quantized **ONNX (xlm-roberta-base)** models to achieve sub-100ms CPU inference for real-time customer sentiment classification.

E-Commerce Customer Sentiment Analyzer | Data Analyst & NLP Engineer 2026

Project Overview: *Large-scale web scraping and sentiment analytics pipeline converting unstructured customer feedback into strategic business intelligence dashboards.*

- Programmed automated Python web scrapers leveraging internal APIs to collect and structure tens of thousands of customer reviews from leading platforms (Tiki).
- Executed extensive text preprocessing pipelines (Vietnamese tokenization, stopword filtering, slang normalization) for high-accuracy sentiment modeling.
- Engineered interactive **Power BI dashboards** to visualize sentiment distributions, product trends, and actionable feedback insights for business strategy.

EDUCATION & ACADEMIC BACKGROUND

Thang Long University

Hanoi, Vietnam

Artificial Intelligence

Sep 2024 – Present (Expected Jun 2028)

• **Cumulative GPA: 3.5 / 4.0**

• **Relevant Coursework:** Data Structures & Algorithms, Machine Learning Foundations, Deep Learning, Applied Statistics & Probability, Database Management Systems (DBMS), Multimodal AI Systems.

TECHNICAL SKILLS & COMPETENCIES

• **Programming Languages:** Python (Advanced / Production-ready), SQL (PostgreSQL, SQL Server), JavaScript / TypeScript, Bash.

• **AI, ML & Frameworks:** PyTorch, LangGraph, CLIP, Hugging Face Transformers, ONNX Runtime, Scikit-learn, Vector DBs (ChromaDB, Supabase PGVector).

• **Backend & Web Engineering:** FastAPI, Next.js (App Router), Streamlit, Redis Queue, Supabase Storage, RESTful APIs, Docker, Git / GitHub Actions CI/CD.

• **Core Methodologies:** Retrieval-Augmented Generation (RAG), Agentic AI & HITL Workflows, Multimodal Vision Routing, Sentiment Analysis, IoT Telemetry, Data Preprocessing.